Problem 16.5

Calculate the accumulated value immediately after the last payment of a 20-year annuity due of annual payments of \$500 per year. The annual effective interest rate is 7%.

Solution

For a 20-year annuity due, the payments are made at the beginning of each year. The accumulated value one period after the last payment would be

$$500\ddot{s}_{\overline{20}|}.$$

However, the question asks for the accumulated value immediately after the last payment, so we discount this value back one year:

$$AV = 500\ddot{s}_{\overline{20}|}v.$$

Since

$$\ddot{s}_{\overline{n}|} = (1+i)s_{\overline{n}|}$$

and $v = (1+i)^{-1}$, we have

$$\ddot{s}_{\overline{20}|}v = s_{\overline{20}|}.$$

Thus,

$$AV = 500s_{\overline{20}|}.$$

Using $i = 0.07$,

$$\begin{aligned} AV &= 500\left(\frac{(1.07)^{20}-1}{0.07}\right) \\ &\approx 20497.7462. \end{aligned}$$

Hence, the accumulated value immediately after the last payment is

$$\boxed{\$20{,}497.75}.$$